ChatGPT

Normal Distributions Comparison

2/11/2026, 1:01:33 AM

Source: https://chatgpt.com/c/698c19fb-cbcc-832a-a0d9-2609c6fe119d

Two different normal distributions could have different means and the same standard deviation.

Question options:
Unselected
True
Selected
False

The correct answer is **True** ✅.

Here's why in simple terms:

- A **normal distribution** is defined by two parameters: the **mean (μ)** and the **standard deviation (σ)**.
- Two distributions can have **different means** (so their centers are at different points) but the **same standard deviation** (so they have the same spread).

For example:

- Distribution A: μ = 10, σ = 2
- Distribution B: μ = 15, σ = 2

They are both normal, different in mean, but same spread.

So it's definitely **True**.

Do you want me to draw a quick diagram showing this?

The following fact is not true about standard normal distribution curve:

Question options:
Unselected
The area under the curve is

Selected
The mean is
and the standard deviation is

Unselected
Can be used with any data distribution

Unselected
Can be used to find the probability of a random variable is less than a constant

The statement that is **not true** about the standard normal distribution is:

"Can be used with any data distribution" ❌

Here's why:

- A **standard normal distribution** specifically has **mean μ = 0** and **standard deviation σ = 1**.
- The **area under the curve is 1** ✅
- It **can be used to find probabilities** for a random variable being less than a constant ✅
- But it **cannot be directly used for any data distribution**. Only data that is **normal (or transformed to normal)** can use the standard normal table directly.

So the correct choice is:

Can be used with any data distribution

If you want, I can quickly explain **how you can use it with non-standard normal data** too—it's actually pretty common.

* A **standard normal distribution** specifically has **mean μ = 0** and **standard deviation σ = 1**.
how?

Ah, got it! Here's how you can use the **standard normal distribution** even if your data doesn't have μ = 0 and σ = 1:

You **convert your data to a standard normal form** using **z-scores**.

The formula is:

$$z = \frac{X - \mu}{\sigma}$$

Where:

- X = the value from your data
- μ = mean of your data
- σ = standard deviation of your data
- z = the corresponding value in **standard normal distribution**

Steps:

1. Take your data value X.
2. Subtract the mean μ.
3. Divide by the standard deviation σ.
4. Use the resulting z to find probabilities from the standard normal table.

Example:

- Suppose your data: μ = 10, σ = 2, and X = 14.
- Convert to z: $z = \frac{14-10}{2} = 2$
- Now, you can look up z = 2 in the standard normal table to get the probability.

✅ This way, **any normally distributed data** can be "standardized" and used with the standard normal distribution.

If you want, I can **draw a quick diagram showing this transformation**—it makes it super clear. Do you want me to?

mean = 0 ?

Yes! After standardization, the **mean becomes 0**.

Here's why:

- Original data: mean = μ, standard deviation = σ
- Standardized data: $z = \frac{X-\mu}{\sigma}$

If you take the **mean of all z-values**:

$$\text{mean of } z = \frac{\text{mean of } X - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = 0$$

So **all z-scores are centered around 0**, and the **spread becomes 1** (standard deviation = 1).

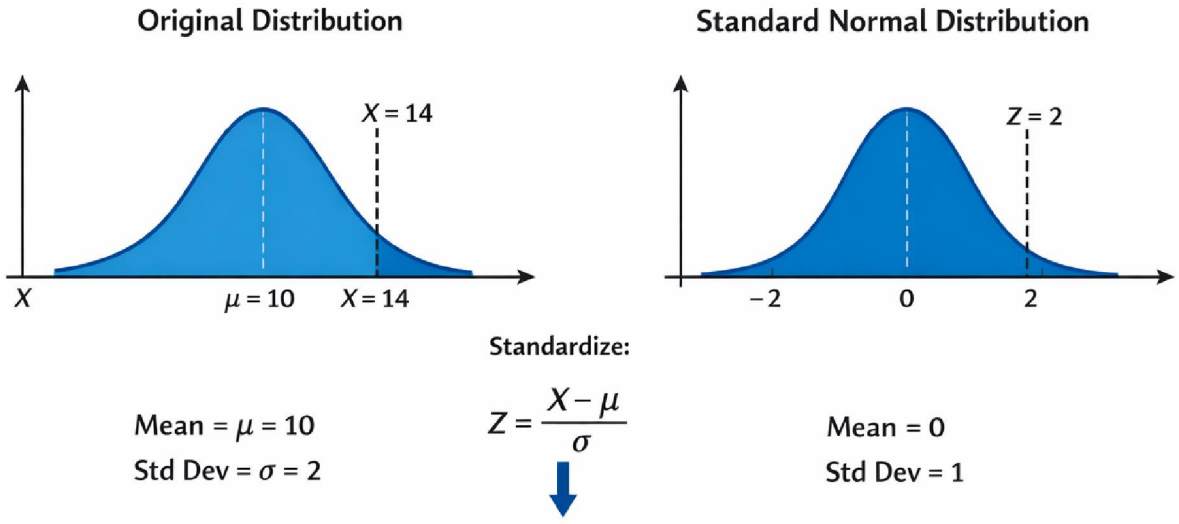
That's why the standard normal distribution always has:

- Mean = 0
- Standard deviation = 1

Basically, you "shift" the data to center at 0 and "scale" it to make the spread 1.

If you want, I can **show a quick diagram of original vs standardized distribution**—it makes it very easy to see. Do you want me to do that?

explain me with geaph


Original Distribution
Standard Normal Distribution
X = 14
X
μ = 10
X = 14
Z = 2
−2
0
2
Standardize:
Mean = μ = 10
Std Dev = σ = 2
Z = (X − μ) / σ
Mean = 0
Std Dev = 1